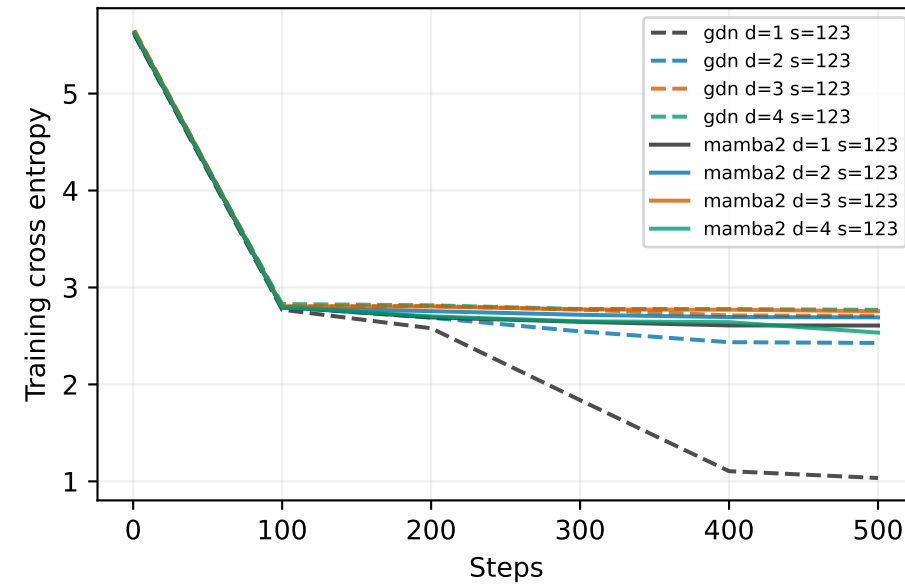
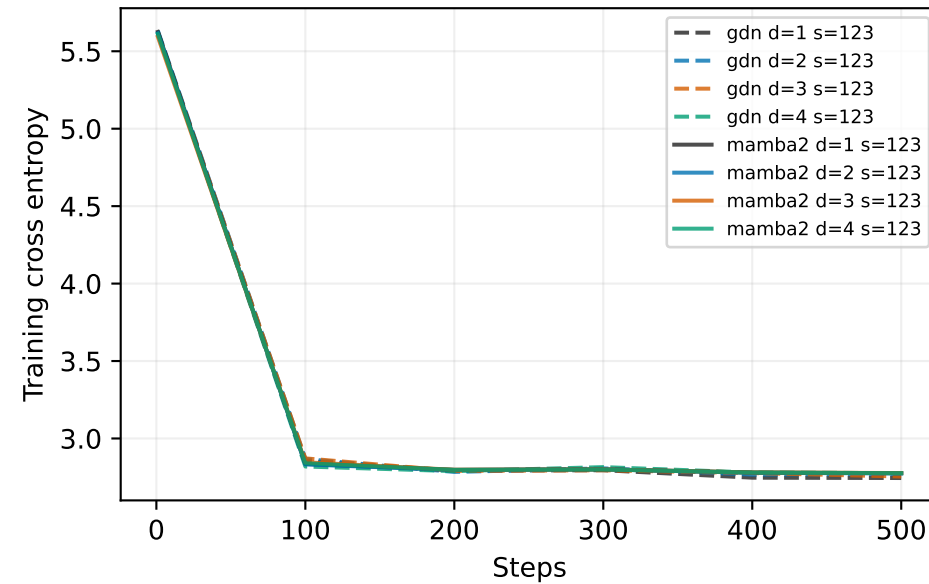


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main500: state, width 64
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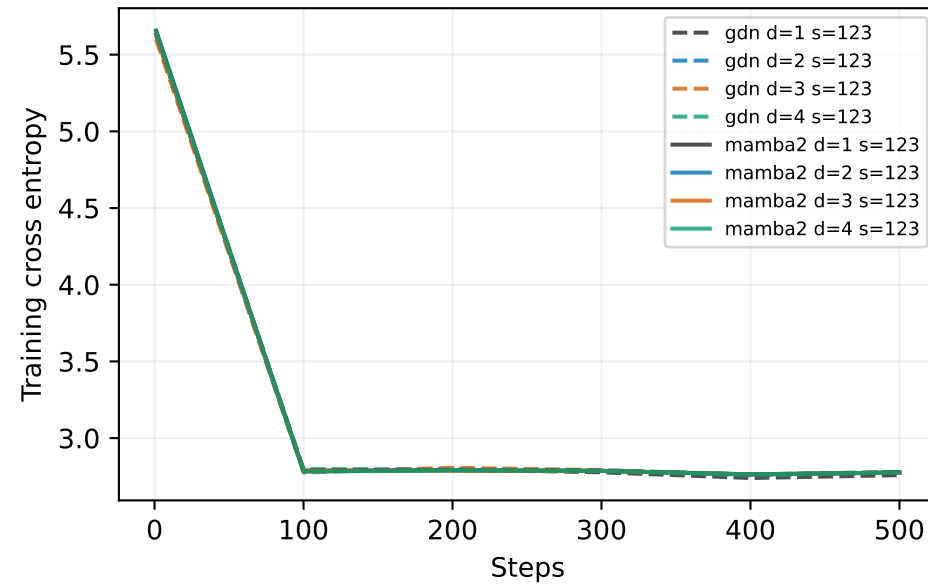
Load 4



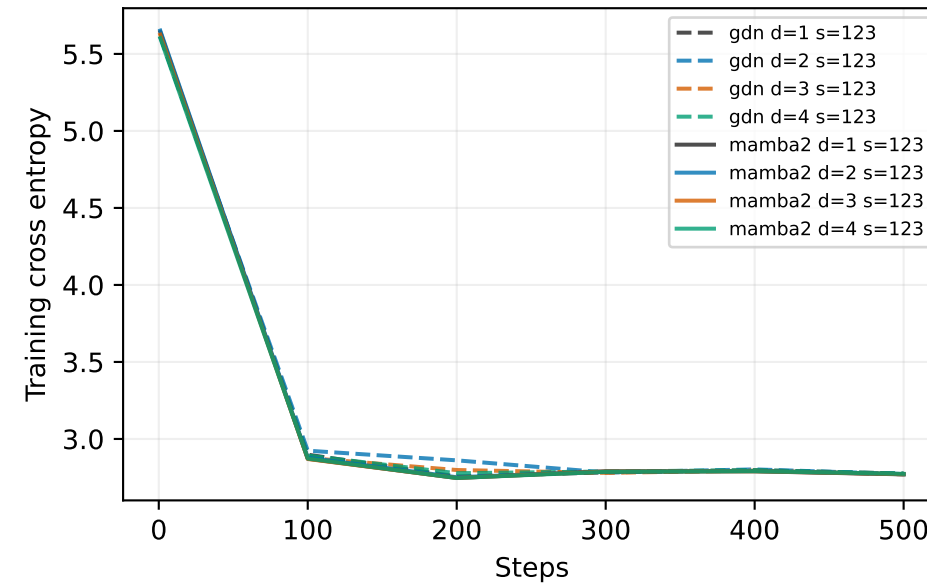
Load 8



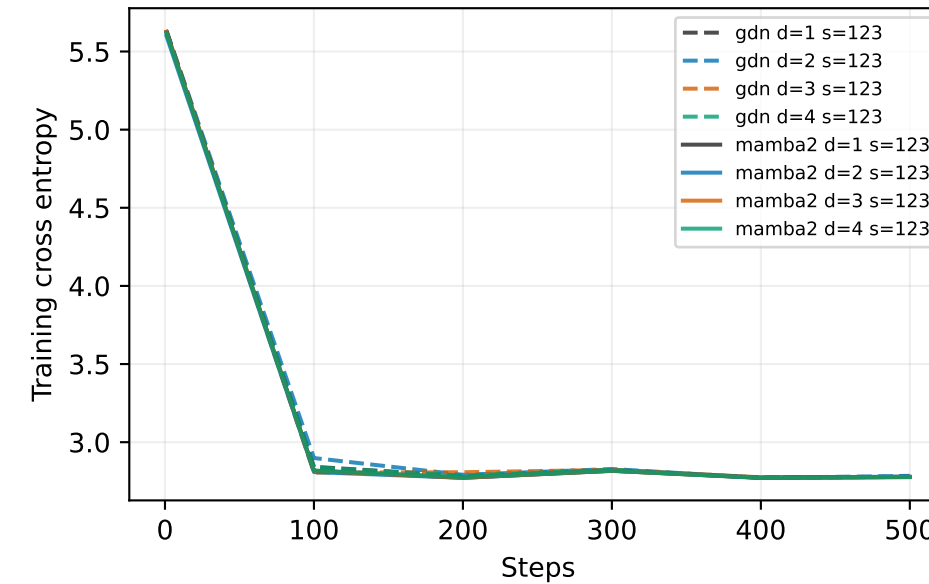
Load 16



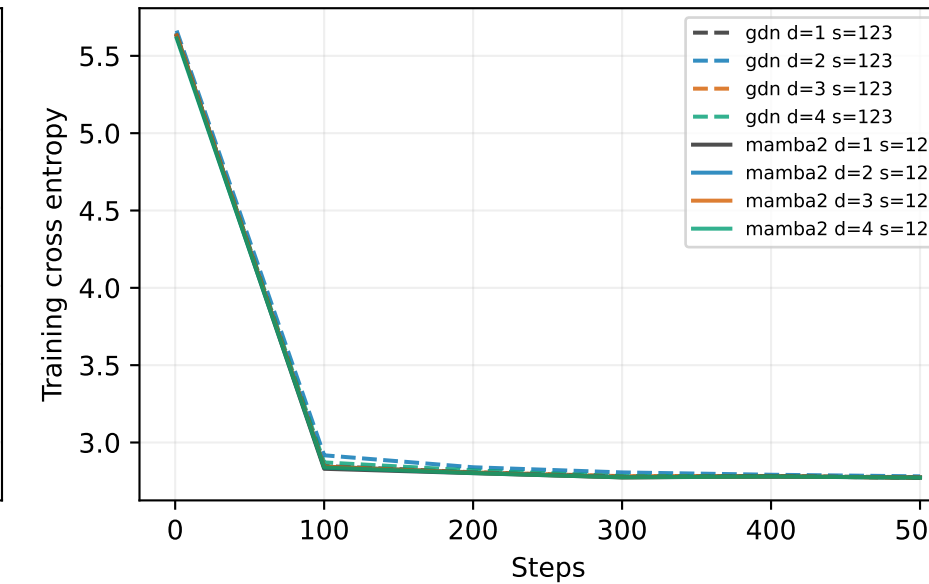
Load 32



Load 64



Load 128



Steps	Ours (dashed black)	Ours w/o ViT (dashed blue)	ViT (solid green)	ViT w/o ViT (dashed orange)	ViT w/o ViT w/o ViT (dotted grey)
0	~5	~5	~5	~5	~5
100	~10	~8	~8	~8	~5
200	~15	~10	~12	~8	~5
300	~40	~15	~18	~8	~5
400	~65	~18	~19	~10	~5
500	~70	~18	~19	~9	~5

Steps	Baseline (grey dotted)	Baseline + Adv. (green dashed)	Baseline + Adv. + 100% Adv. (orange dashed)	Baseline + Adv. + 10% Adv. (blue dashed)	Baseline + Adv. + 1% Adv. (brown dashed)
0	7	3	2	2	2
100	7	7	6	6	7
200	7	6	6	6	7
300	7	6	6	6	7
400	7	6	6	6	8
500	7	6	6	6	10

Validation accuracy (%)

Steps

Steps	Baseline (grey dotted)	Single layer (red solid)	Two layers (orange solid)	Three layers (green dashed)	Four layers (black dashed)
0	7	72	75	2	2
100	7	68	70	5	5
200	7	65	68	6	6
300	7	62	65	6	7
400	7	60	63	6	8
500	7	60	63	6	9

Steps	Baseline (dotted grey)	Single Layer (dashed green)	Two Layers (solid blue)	Three Layers (solid red)
0	~7%	~0%	~4%	~5%
100	~7%	~6%	~7%	~7%
200	~7%	~5%	~6%	~6%
300	~7%	~5%	~6%	~6%
400	~7%	~5%	~6%	~6%
500	~7%	~5%	~6%	~7%

Steps	Llama2 (Red)	Llama2+LoRA (Green)	Llama2+LoRA+QAT (Blue)
0	2.0	4.0	6.0
100	5.0	6.0	7.0
200	5.0	6.0	7.0
300	5.0	6.0	7.0
400	5.0	6.0	7.0
500	5.0	6.0	7.0

Steps	Baseline (dotted)	Single Layer (dashed)	Two Layers (solid)	Three Layers (dash-dot)	Four Layers (long-dashed)
0	6.5	0.0	4.0	4.5	4.5
100	6.5	6.0	6.5	6.5	7.0
200	6.5	6.0	6.5	6.5	7.0
300	6.5	6.0	6.5	6.5	7.0
400	6.5	6.0	6.5	6.5	7.0
500	6.5	6.0	6.5	6.5	7.0